Step 4 Result 2.0: Independent Temperature and Field Trends

1 Purpose of the test

This report applies the Step 4 validation protocol to the Result 2.0 workflow. The test separates a dimerisation/lattice channel, which follows $\delta(T, B)$, from a scalar spin-correlation proxy $C_1(T, B)$ that remains finite above the spin-Peierls transition.

2 Physical model

The active mixing term is

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q^{\text{eff}} Q L_{BD}, \tag{1}$$

with

$$V_{BD}^{\text{static}} = V_0 + \lambda_\delta \delta(T, B) + \lambda_C C_1(T, B), \qquad g_Q^{\text{eff}} = g_{Q0} \delta(T, B)/\delta_{\text{ref}}. \tag{2}$$

The dynamic phonon channel therefore turns off when the dimerisation vanishes.

3 Trend definitions

The four scenarios are the low-temperature lattice reference, the high-temperature lattice-off control with finite C_1, the field-suppressed low-temperature control, and the high-temperature C_1-static control. The frequency convention is rephasing $\omega_1 < 0$, unrephasing $\omega_1 > 0$, and $\omega_3 > 0$ for both components. The current grid is $N_w = 300$ with rephasing $\omega_1 = [-1.1, -0.8]$, unrephasing $\omega_1 = [0.8, 1.1]$, and $\omega_3 = [0.8, 1.1]$ eV.

4 Scenarios and acceptance criteria

Scenario	δ	C_1	g_Q^{eff}	V_{BD}^{static}	A_{reph}
lowT_lattice_dynamic	0.00707107	-0.378667	0.01	0.02	231.909
highT_lattice_off_C1_finite	0	-0.329471	0	0.01	250.111
highB_lattice_suppressed	0	-0.346199	0	0.01	250.111
highT_C1_static	0	-0.329471	0	0.02	240.227

The acceptance criteria are that the low-temperature lattice reference has finite δ and g_Q^{eff}, both high-temperature and high-field lattice-off controls have $\delta = 0$ and $g_Q^{\text{eff}} = 0$ while retaining finite C_1, and the C_1-static control demonstrates that a finite scalar channel can persist without dimerisation.

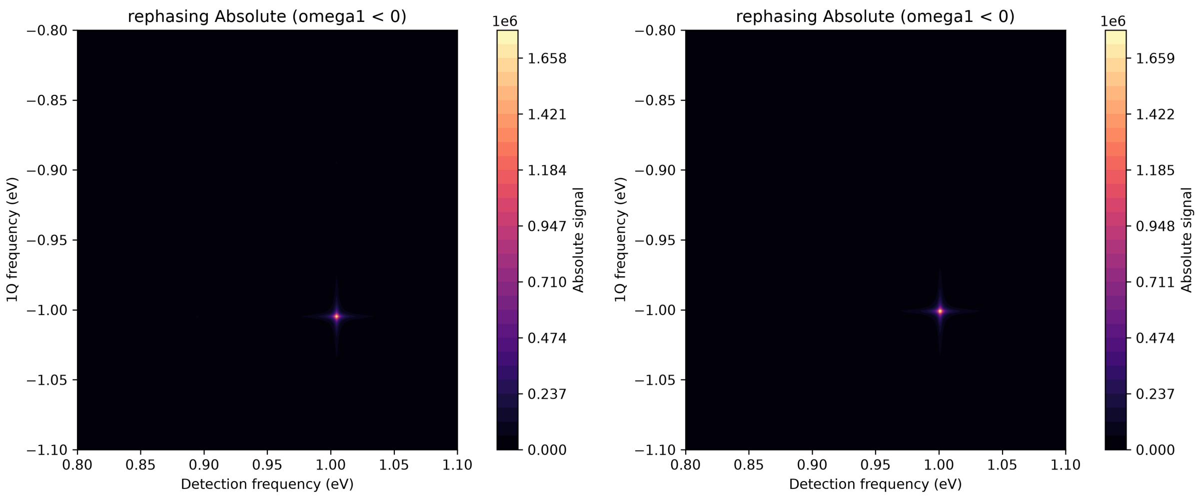

Figure 1: Representative rephasing absolute spectra. Left: low-temperature lattice dynamic reference. Right: high-temperature lattice-off control.

5 Numerical results

The analysis files are `step04__scenario_metrics.csv`, `step04__pairwise_differences.csv`, and `step04__validation_outcome.txt` in the `Analysis/` folder.

6 Validation outcome

The Step 4 production and trend-control checks pass for the generated scenarios. The high-temperature and high-field controls remove the dynamic lattice channel, while the C_1-static case confirms that a persistent scalar contribution should not be interpreted uniquely as evidence of the dimerised phase.

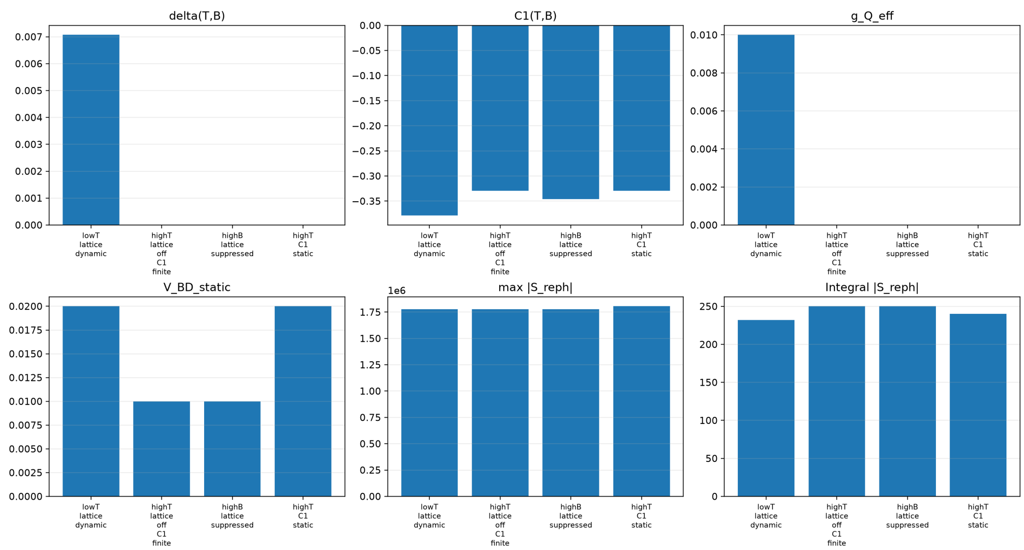

Figure 2: Step 4 trend metrics from the Result 2.0 output files.